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REVIEW

How consumer physical activity monitors could transform human physiology research

Stephen P. Wright,[1] Tyish S. Hall Brown,[2] Scott R. Collier,[3] and Kathryn Sandberg[1]

[1]*Department of Medicine, Georgetown University, Washington, D.C.;* [2]*Department of Psychiatry & Behavioral Sciences, Howard University College of Medicine, Washington, D.C.; and* [3]*Department of Health and Exercise Science, Appalachian State University, Boone, North Carolina*

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Wright SP, Hall Brown TS, Collier SR, Sandberg K. How consumer physical activity monitors could transform human physiology research. *Am J Physiol Regul Integr Comp Physiol* 312: R358–R367, 2017. First published January 4, 2017; doi:10.1152/ajpregu.00349.2016.—A sedentary lifestyle and lack of physical activity are well-established risk factors for chronic disease and adverse health outcomes. Thus, there is enormous interest in measuring physical activity in biomedical research. Many consumer physical activity monitors, including Basis Health Tracker, BodyMedia Fit, DirectLife, Fitbit Flex, Fitbit One, Fitbit Zip, Garmin Vivofit, Jawbone UP, MisFit Shine, Nike FuelBand, Polar Loop, Withings Pulse O_2, and others have accuracies similar to that of research-grade physical activity monitors for measuring steps. This review focuses on the unprecedented opportunities that consumer physical activity monitors offer for human physiology and pathophysiology research because of their ability to measure activity continuously under real-life conditions and because they are already widely used by consumers. We examine current and potential uses of consumer physical activity monitors as a measuring or monitoring device, or as an intervention in strategies to change behavior and predict health outcomes. The accuracy, reliability, reproducibility, and validity of consumer physical activity monitors are reviewed, as are limitations and challenges associated with using these devices in research. Other topics covered include how smartphone apps and platforms, such as the Apple ResearchKit, can be used in conjunction with consumer physical activity monitors for research. Lastly, the future of consumer physical activity monitors and related technology is considered: pattern recognition, integration of sleep monitors, and other biosensors in combination with new forms of information processing.

activity tracker; accelerometer; fitness trackers; wearable device

CONSUMER PHYSICAL ACTIVITY MONITORS are widely used by the public to measure daily step counts. As of November 11, 2016, 414 models of consumer physical activity monitors were available on the market (http://fitness-trackers.specout.com). A survey (released January 6, 2015) of 5,000 U.S. adults reported that 1 out of 10 people wears a consumer physical activity monitor (www.livescience.com/49400-fitness-tracker-smartwatch-survey.html) with an expected 11% increase in use by the end of 2016 (www.healthcareitnews.com/blog/wearable-activity-tracking-device-purchasing-expected-grow-11-percent-2016).

The National Institutes of Health (NIH) defines physical activity as any body movement that uses your skeletal muscles and requires more energy than resting (13). Step count is one measure of physical activity and one that is positively correlated with health outcomes. Using pedometers, Dwyer et al. (22) showed that participants with baseline step counts that reached or exceeded 10,000 daily steps had a 46% lower risk of death over the next 10 years. In contrast, a sedentary lifestyle is associated with a myriad of adverse health outcomes, including increased risk for disease incidence, mortality, and hospitalization (8). Thus, researchers studying human physiology and pathophysiology have an enormous interest in measuring physical activity because of its profound impact on health and disease.

This review, intended for a general physiology audience, focuses on the new research opportunities and potential advantages of wearable consumer physical activity monitors for measuring activity and use as a behavioral intervention. In this context, "wearable" refers to monitors that can be worn on the wrist or clipped to an individual's clothing and does not include smartphones. The limitations and challenges posed by using these devices as a research tool are also addressed, as well as a discussion of the next generation of consumer

Address for reprint requests and other correspondence: S. P. Wright, 232 Building D, Georgetown Univ., 4000 Reservoir Rd., NW, Washington, DC 20057 (e-mail: spw44@georgetown.edu).

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physical activity monitors and their integration with other wearable devices for monitoring health.

METHODS

Searches and Search Criteria

Literature searches were conducted in Ovid MEDLINE using terms or phrases in parentheses in the keyword search field with the "map term to subject heading" field deselected. Terms or phrases were also searched in PubMed MEDLINE using the basic search option for text ("phrase" [Text]).

Clinical trials were searched through ClinicalTrials.gov, which is a "registry and results database of publicly and privately supported clinical studies of human participants conducted around the world" (https://clinicaltrials.gov/ct2/search/advanced). Search terms were used in conjunction with "All Studies" being selected under the advanced search terms for "Recruitment", "Study Type", "Study Results" and "Gender". All other search fields were left empty.

A search of NIH-funded grants was conducted through the NIH RePORTER, which is an "electronic tool that allows users to search a repository of NIH-funded research projects and access publications and patents resulting from NIH funding" (https://projectreporter.nih.gov/reporter.cfm/). The "Projects" field was selected, and key words were searched through the "TEXT SEARCH" field. The "Publications" and "News" fields were deselected as were the "Limited to: Project Title, Project Terms, or Projects Abstracts" fields. All search fields under "RESEARCHER AND ORGANIZATION," "PROJECT DETAILS," and "ADDITIONAL FILTERS" were left empty.

A Note about Nomenclature

Consumer physical activity monitors are referred to variously (in academic and nonacademic writings), as "activity monitors," "activity trackers," "exercise monitors," "exercise trackers," and "step trackers," and sometimes these phrases are preceded by "consumer," "commercial," "popular," and other descriptors. A search of the biomedical literature shows that "activity monitor" is the most commonly used term to describe devices used to conduct physical activity measurements (Table 1). "Monitor" is also the preferred term when describing ecological assessments of heart rate (HR) and blood pressure. In this review, we use the term "consumer" to distinguish these wearable devices from "research-grade" physical activity monitors. When we refer to both kinds collectively or when it's appropriate to make no distinction, we use "physical activity monitors" or simply "activity monitors."

An accelerometer is a common electronic component inside an activity monitoring device. Many researchers have used "accelerometer" as shorthand for the physical activity monitor (especially of the research-grade variety). We do not use this shorthand here, in order to be clear about the distinction between the working core of the device (the accelerometer) and the device itself (including additional electronics, the display, housing, and strap).

Table 1. *Frequency of descriptor cited in the biomedical literature*

Phrase	Number of Citations in Ovid MEDLINE	Number of Citations in PubMed MEDLINE
Activity monitor	955	1,078
Activity tracker	10	42
Fitness monitor	3	0
Fitness tracker	8	9
Exercise monitor	1	0
Exercise tracker	0	0
Step tracker	0	0

Basic searches were conducted in Ovid MEDLINE using the "phrase" in the keyword search field with the "map term to subject heading" field deselected and in PubMed MEDLINE using the basic search option for text ("phrase").

Opportunities to Continuously Assess Physical Activity Under Real-Life Conditions (Ecological Assessment) in the Context of Human Physiology and Pathophysiology

Research-grade physical activity monitors are designed to store all data on the device itself. Thus far, device storage capacity has limited data collection to a period of several weeks (or at most a few months if at a low sample rate, in the case of the most popular research-grade monitor, the Actigraph GT3X) (1a). While ongoing miniaturization of memory will likely expand the data collection period in the future, currently, participants need to return the device to enable investigators to extract the data from that specific period. One of the major advantages of consumer physical activity monitors is that the data are collected and transferred continuously (or at frequent intervals) to a smartphone or website. Thus, consumer monitors offer the possibility of continuous assessment of physical activity over months or even years.

Continuous ecological assessment of physical activity provides an opportunity to markedly advance our understanding of the impact of frequency, duration, quality, and intensity of physical activity on human physiology and pathophysiology. Improved reference data will refine our evidence-based recommendations for physical activity guidelines, which can be used by the medical community and public health officials to communicate more accurately and effectively with the public to encourage and support healthy behavior change.

The company Fitbit has only been in existence since 2007, but it is the leading producer of consumer physical activity monitors, with 79% market share by sales (https://www.npd.com/wps/portal/npd/us/news/press-releases/2016/year-over-year-wearables-spending-doubles-according-to-npd/). Searches conducted and updated to November 10, 2016, show that Fitbit is being used more and more in biomedical research. One hundred and twenty-seven clinical studies using Fitbit devices are listed in clinicaltrials.gov (Fig. 1*A*). Forty-eight have been completed, while 79 are ongoing. Some studies have already published their findings. Forty papers cited in Ovid MEDLINE report use of a Fitbit device (Fig. 1*B*). NIH is now funding research that uses consumer physical activity monitors, and the NIH RePORTER shows the number of grants using Fitbit is rapidly increasing (Fig. 1*C*). The explosion in publications and exponential use of Fitbit devices in clinical trials and NIH-funded research illustrates the growing appreciation of the research potential these consumer devices present to biomedical research. While Fitbit is the most popular device used in the studies, a search of clinicaltrials.gov (number in brackets is the number of trials citing study use) as of November 11, 2016 for Jawbone [7], Apple watch [4], Garmin [4], Polar [4], Microsoftband [2], Misfit [2], Nike [1], Withings [1] and Xiaomi [1] illustrates that various brands of consumer physical activity monitors are being used in clinical trials.

Consumer Physical Activity Monitors: Types of Use

Analysis of Fitbit use in clinicaltrials.gov as of November 11, 2016, shows that out of 127 total studies, the majority of these are interventional, conducted in adults with targeted enrollments of less than 100 (Table 2). The diversity in primary outcome measures is illustrated in Fig. 2*A* and demonstrates that physical activity is the most common primary outcome. Biomarker measurement is the second most common primary outcome and encompasses an array of health issues with the majority centering on metabolic function (e.g., HbA1c and fasting glucose) and cognitive or neurological health (e.g., executive function and migraines) (Fig. 2*B*). Examples of current use are discussed below. The ascribed clinical trial number in clinicaltrials.gov is listed in brackets.

Measuring physical activity to validate experimental paradigms. The European Institute of Oncology is using the Garmin Vivofit

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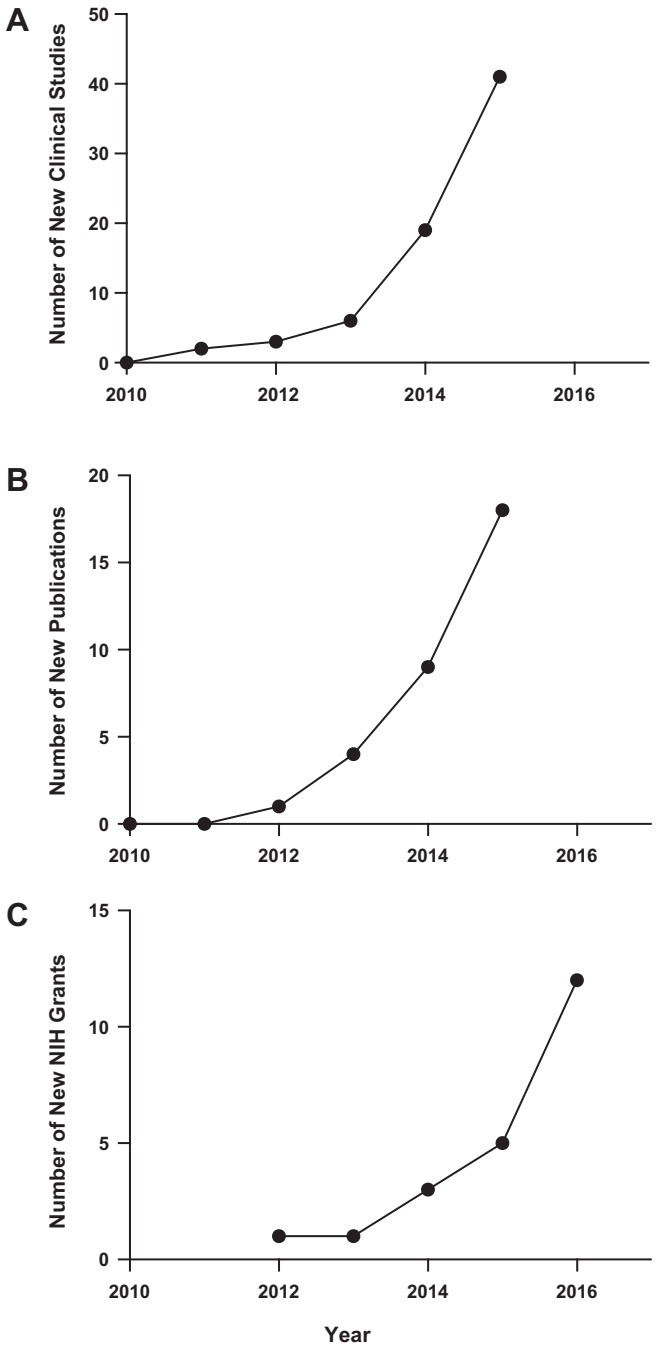


Fig. 1. Rapid growth of research involving Fitbit activity monitors. The number per year of clinical studies listed in clinicaltrials.gov (*A*), publications in the Ovid Medline database that cite Fitbit in their abstracts (*B*), and grants in the NIH RePORTER database (*C*), as of November 9, 2016. The "Full Text View," "Tabular View," and "Study Results" were examined for each clinical trial result to ensure relevance and to avoid duplicates of study entries. The year recruitment began for each selected study was noted. For each result in Ovid MEDLINE, the title and abstract were examined to ensure relevance and to avoid duplication, and the year of publication was noted. The project number, project title, and abstract were examined for each result found in the NIH RePORTER database to ensure relevance and avoid duplication and the first year of project funding was noted.

activity and sleep monitor to determine whether an 8-wk exercise program will improve quality of life in women with a personal history of breast cancer [NCT02637765]. In this study, all subjects were provided with the Vivofit. The intervention in this study is the 8-wk exercise program, while the Garmin Vivofit is the tool used to validate that the women in the 8-wk exercise program are more active than the control group. The primary outcome is quality of life.

The University of Minnesota is using data from Fitbit activity and sleep monitors to determine the effect of exercise intensity on smoking cessation in young adult smokers with the primary outcome being pulmonary health [NCT02422914]. In this study, the Fitbit device is used as the tool to validate that participants in the high-intensity exercise group are exercising more intensely than the lower-intensity exercise group.

Comparing physical activity to physical fitness. Physical fitness is a more complex concept than physical activity and is defined as a set of health- or skill-related attributes like body composition, muscular fitness, endurance, and cardiorespiratory fitness (13). Repeated bouts of physical activity contribute to a higher level of physical fitness (23). In fact, it is well known that exercise training exerts beneficial cardiopulmonary effects in as few as 4 weeks with moderate-intensity exercise sustained for 20–30 min, three times per week, for both diseased and healthy populations (1, 15).

While physical activity is a building block of physical fitness, it is not a direct measure of physical fitness, and exercise physiologists continue to debate the relative importance of physical activity vs. physical fitness in predicting health outcomes. A recent study of 206 elderly individuals, who wore a uniaxial pedometer continuously for one year suggests that physical activity is more closely associated with arterial stiffness than estimates of aerobic fitness determined by a standard walking test and maximal walking speed (4). In contrast, other studies suggest aerobic fitness is the major predictor of arterial stiffness in the elderly (3, 46). Consumer physical activity monitors provide a new tool for weighing in on this debate. Studies comparing ecological assessments of long-term patterns of physical activity duration, frequency, and intensity with measures of $V_{O_{2max}}$, static and dynamic muscle contraction performance for various muscle groups, flexibility, balance, agility and coordination—as well as percent body fat and lean body mass in the context of health, aging, and disease—will markedly advance our understanding of human physiology and pathophysiology. For example, a clinical trial conducted by New York Institute of Technology is evaluating the association of activity (determined by a Fitbit Flex wrist-worn monitor) with a measure of fitness ($\dot{V}_{O_{2max}}$) as a function of a wellness intervention [NCT02778009].

Measuring physical activity to investigate associations between behavior and activity. A University of South Carolina study is using the MisFit wrist-worn activity and sleep monitor to compare

Table 2. *Characteristics of clinical trials in clinicaltrials.gov citing use of Fitbit physical activity monitors*

Study Type	Number of Clinical Trials
Observational	11
Interventional	116
Total	127
Participant age	
Children only (<14 yr old)	4
Children to young adult (<24 yr old)	10
Adults only (>18 yr old)	107
All ages	6
Estimated enrollment	
1–99	88
100–999	34

A search of the term "Fitbit" was conducted in clinicaltrials.gov. The table shows the summary of study type, eligibility criteria under recruitment information, and the estimated enrollment.

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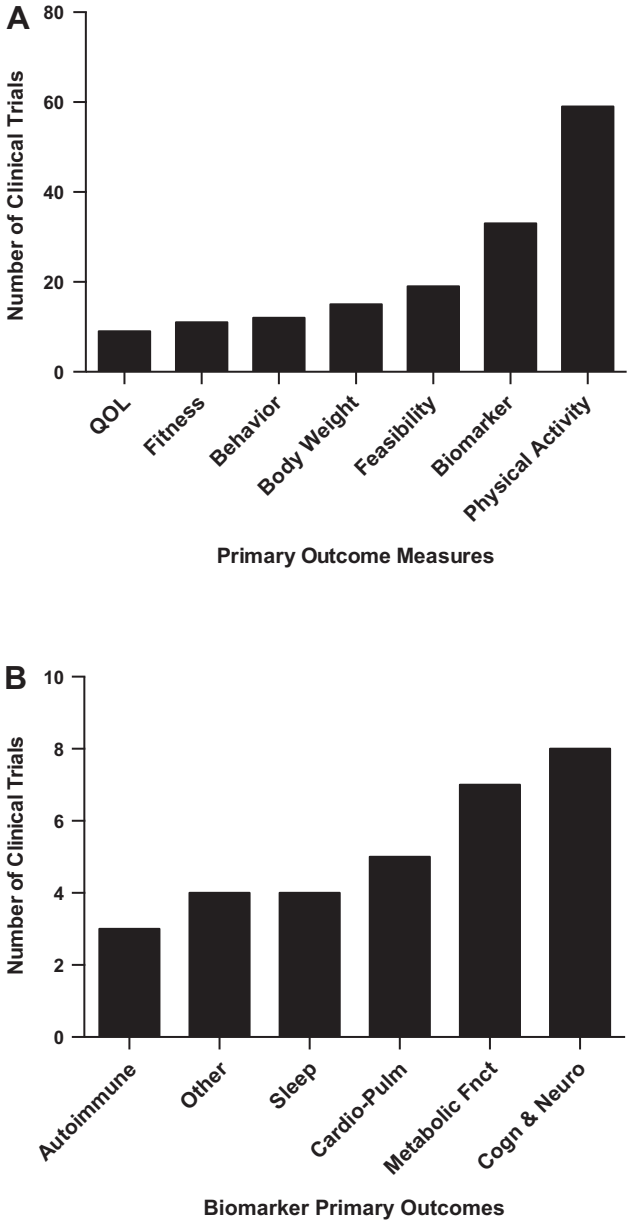


Fig. 2. Primary outcome measures of studies involving Fitbit activity monitors. The number of clinical trials as of November 10, 2016, in clinicaltrials.gov citing use of Fitbit for various primary outcomes, including quality of life (QOL), fitness (e.g., exercise capacity and walking distance), behavior (e.g., medical adherence, diet, abstinence, smoking), weight (e.g., body mass index, waist circumference, lean and fat muscle mass), feasibility (e.g., recruitment, retention, adherence, satisfaction), biomarker (e.g., blood pressure, heart rate, sleep, fasting insulin, HbA1c) and physical activity (e.g., step count, moderate to vigorous physical activity) (*A*) and biomarker primary outcomes, including autoimmune, sleep, cardio-pulmonary (Cardio-Pulm), metabolic function (Metabolic Fnct) and cognition and neurological (Cogn and Neuro) indicators (*B*), based on primary outcome categories listed in clinicaltrials.gov.

activity levels between people on vegan vs. omnivorous diets [NCT02417480].

Measuring physical activity to predict health outcomes. Data from these devices (e.g., daily steps) are being used in naturalistic studies to predict outcomes. Arkansas Children's Hospital is studying whether Fitbit Charge HR wristband data will predict lung function in young adults with asthma [NCT02556567], and Wake Forest University is conducting a feasibility study to ultimately assess whether Fitbit Zip data can predict surgical complications after abdominal cancer surgery [NCT02356471]. The hypotheses for these studies is that individuals who are more active will have better lung function (asthma symptoms are the primary outcome) and fewer surgical complications, respectively. Researchers at L'Université Paris-XIII-Nord studied 19,000 adult users of Withings Pulse physical activity monitors and blood pressure monitors and determined that daily walking is associated with decreased systolic blood pressure in both cross-sectional and longitudinal analyses (11).

Measuring physical activity to inform treatment decisions. Cedars-Sinai Medical Center in Los Angeles is using the wrist-worn Fitbit Charge HR in adults with advanced cancer to assess fitness for chemotherapy [NCT02659358]. The hypothesis is that a threshold of activity will be discovered that can be used to determine who can handle chemotherapy. Similar studies are using devices to monitor individuals at home so that low activity will trigger a healthcare intervention (see discussion of new Apple framework called CareKit under *"Opportunities to use smartphone apps and platforms such as the Apple ResearchKit in conjunction with consumer physical activity monitors"* below).

Feedback from consumer physical activity monitors as a behavioral intervention to increase physical activity for disease prevention. Consumer physical activity monitors provide feedback to the user regarding activity on the device itself, through an app on a user's smartphone, and/or through a website. A randomized trial in women demonstrated that wearing a Fitbit One clip-on monitor increased physical activity during a 16-wk period compared with a pedometer control group (12). The authors concluded that feedback generated by the Fitbit on its website was more effective at motivating women to engage in physical activity than the simple step information provided by the pedometer. In this study, the the primary outcome was physical activity determined by the concurrently worn research-grade ActiGraph GT3X+. In another study, user feedback was associated with an increased likelihood to log greater than 10,000 steps per day over a 90-day study period compared with those who did not receive this feedback (35). Participants could log their steps either by using the 10,000 Steps website or through their smartphone app.

Many consumer physical activity monitors use behavior change techniques to motivate the user, with goal-setting, self-monitoring, and feedback being the most prevalent (42). Setting goals and getting feedback on past and current activity is commonly employed, and there are often opportunities for social interactions that can provide support, competition, and other motivational strategies. Investigators studying behavior change techniques on physical activity can compare motivational strategies in conjunction with consumer monitors.

An ongoing study at Duke University is comparing the mobile health coaching application Vida with a live health coach in conjunction with Fitbit feedback and motivational techniques [NCT02431546]. A primary outcome measure in this study is physical activity. In this case, Fitbit is both the outcome measure and part of the intervention. The vast majority of these behavioral intervention studies involve education and behavioral reinforcement techniques like alarms, texting, and coaching (42).

Feedback from consumer physical activity monitors as a behavioral intervention to lose weight and reduce body mass index to improve health outcomes. Fitbit Ultra monitors are also being used in one arm of a comparative trial [NCT01693250] at the University of California, San Francisco, aimed at preventing diabetes in adolescents. The primary outcome is change in body mass index after 3 mo and the Fitbit Ultra is the intervention. Lancaster General Hospital in Pennsylvania is using a wrist-worn Fitbit as the sole intervention in a study [NCT02381262] of patients after vertical sleeve gastrectomy to determine whether wearing Fitbits will improve weight loss over a 12-wk period. The primary outcome is weight loss. The intervention is wearing the Fitbit device.

Feedback from consumer physical activity monitors as a behavioral intervention to attenuate disease progression. Most current clinical trials involving these consumer physical activity monitors use

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them as a motivational intervention to promote physical activity and weight management with the goal of improving health outcomes in populations with a range of adverse conditions: childhood obesity, diabetes, cystic fibrosis in adolescents, recovering alcoholics, peripheral artery disease, and knee osteoarthritis. For example, Children's Hospital in Boston is using the Fitbit as an intervention (NCT02700243) to supplement standard counseling in young adult cystic fibrosis patients. The primary outcome is an exercise test to assess fitness. The intervention is wearing a Fitbit. The hypothesis is that wearing a Fitbit will increase the level of exercise and, thereby, improve lung function.

Limitations and challenges associated with using consumer physical activity monitors in biomedical research. Accelerometry is an objective method of quantifying physical activity and energy expenditure during the day, as well as at night (47). Integrated chip sensors typically comprise a capacitive micro-electro-mechanical, piezo-electric, or piezo-resistive element that detects the change in acceleration of a small mass in the sensor. Raw data from the accelerometers (housed in both research-grade physical activity monitors and consumer physical activity monitors) are utilized in algorithms to count steps and sometimes to calculate parameters for other activities (or inactivity).

Although the hardware and especially the algorithms vary, the newer research-grade physical activity monitors and consumer physical activity monitors generally employ a triaxial accelerometer (51), which records acceleration vectors in three planes, and this represents a significant advance over devices using a uniaxial accelerometer in their ability to calculate energy expenditure (49).

Algorithms (which are continually improving) interpret this data to estimate the frequency, duration, and intensity of physical activity-like steps, flights of stairs, or energy expenditure. One method involves using vertical and anterior-posterior (front and back) accelerations to calculate step frequency and determine walking patterns while mediolateral (side-to-side) accelerations predict stride frequency and can also be used to assess abnormalities in gait (34).

Unknown algorithms. The algorithms used in consumer physical activity monitors to determine steps taken, distance traveled, and energy expenditure are typically not shared with researchers due to proprietary concerns. Not only is the investigator blinded to these algorithms, most companies are actively engaged in research and development and update these algorithms and motivational techniques, according to their own discretion and time frame, which poses a challenge in comparing data among participants who are recruited at different times and also for analyzing longitudinal data from the same participant. Not only might companies change their algorithms without prior notice, they may not reveal this fact afterward, so researchers might not even know about this potential discontinuity in their data over time, which could wreak havoc on the integrity of intersubject and intrasubject data. In addition to relatively frequent firmware updates that may change the algorithms, companies continually upgrade their devices, making older models obsolete. This discontinuation of models could pose a problem for a large study with rolling enrollment, in addition to researchers not knowing to what extent previous validity studies of a device would apply to its successors. Some companies that work with researchers may be willing to notify them of firmware updates and even provide a conversion, as Withings has promised to do in our study of the effect of continuous ecological assessment of physical activity on body weight and blood pressure in the greater Washington, D.C. Black population.

Another limitation is that there is no established consensus on how to calculate metabolic equivalent of task (MET) thresholds within the consumer physical activity monitor business. MET thresholds are commonly used by exercise physiologists to categorize sedentary (<1.5), light (1.5 to <3 MET), moderate (3 to <6 MET) or vigorous (6 or higher MET) activity (56). No consensus means it will be difficult to compare MET findings among studies using different consumer physical activity monitors.

Establishing validity. Investigators using consumer physical activity monitors will need to establish the validity of these research tools, including unit and value calibration, as well as the concurrent and predictive validity of these devices against a gold standard. Definitions and examples of validity terms are summarized from Bassett et al. (6) and are illustrated in Table 3. A recent systematic review of 22 studies using Fitbit and Jawbone consumer physical activity monitors by Evenson et al. (25) shows that most are as accurate as research-grade activity monitors in counting steps; however, the error markedly increases at slow walking speeds. While the error rate was 1% for Fitbit One (55, 57) for normal walking, the device recorded zero steps for participants walking at 0.3–0.5 m/s (52). Thus, this caveat needs to be taken into account in studies of older adults or people with impaired walking ability, although this issue may be circumvented by ankle placement of the activity monitor (36).

Energy expenditure calculations are based on algorithms and are necessary for quantifying physical activity in a single scalar and for assessing energy balance. Both consumer- and research-grade devices are less accurate in estimating energy expenditure compared with determining the number of steps. Error rates are 10–13% or higher. Lee et al. (40) compared several consumer physical activity monitors by treadmill walking and running using the Oxycon Mobile, a portable metabolic system serving as an indirect calorimeter for energy expenditure measurement, as the gold standard. Most consumer physical activity monitors, including BodyMedia Fit, Fitbit Zip, Jawbone Up, DirectLife, and NikeFuel Band, had absolute error values for energy expenditure (calculated from the average value of the group-level errors), which were similar to the research-grade ActiGraph GT3X+ (12.6%). Furthermore, BodyMedia Fit, Fitbit Zip, and the NikeFuel Band were each within 10% equivalence of the indirect calorimetry estimate.

Sasaki et al. (50) showed that error rates in calculating energy expenditure depend upon the type of physical activity. When the Fitbit Classic was compared with the Oxycon Mobile portable metabolic system, energy expenditure was underestimated for cycling, raking,

Table 3. *Definition of calibration and validity terms*

Term	Definition*	Example
Calibration		
Unit calibration	accuracy of the measurement being conducted	counting number of steps correctly
Value calibration	accuracy of the conversion of the monitor's measurement into other measurement units	converting steps into energy expenditure
Criterion-referenced validity		
Concurrent validity	validity of an assessment compared with a known assessment	Is the number of steps counted on an activity monitor as accurate as steps counted on an Actigraph?
Predictive validity	extent to which a device can predict an outcome or parameter compared to the predictive gold standard device	Does the average number of weekly steps measured on an activity monitor predict cardiovascular disease risk to the same extent as measuring average steps on an Actigraph?

*Summarized from Bassett et al. (6).

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laundry, stairs, and treadmill walking at a 5% grade, while energy expenditure was overestimated for carrying groceries. Bai et al. (5) found that three (Fitbit Flex, Jawbone Up24, and the Nike+ Fuelband SE) out of five (also MisFit Shine and Polar Loop) consumer physical activity monitors tested were roughly as accurate in assessing energy expenditure in varied activities simulating free-living conditions reproduced in the laboratory, as two research-grade physical activity monitors (ActiGraph GT3X+ and BodyMedia Core) using Oxycon Mobile as a gold standard. Not surprisingly, the study found that none accurately estimated energy expenditure in resistance exercise because these monitors are not designed to measure the increased energy cost due to lifting weights.

Less is known regarding consumer physical activity monitors and swimming. Several monitors, including Garmin Vivoactive, TomTom Spark, and Misfit Speedo Shine provided accurate information on duration and distance (total and laps), while fewer (e.g., Garmin Vivoactive and TomTom Spark) provided accurate information on average and maximum stroke rate, in a test performed by wareable.com (https://www.wareable.com/fitness-trackers/big-swimming-tracker-review). No studies to date, however, have analyzed the accuracy of their energy expenditure calculations for swimming.

Thus, investigators will need to test the interunit and intraunit precision and reproducibility and conduct quality control measures using gold standards to ensure these consumer physical activity monitors are accurate throughout the study.

Special Challenges Associated with Long-Term Use of Consumer Physical Activity Monitors

Basic adherence. According to a 2014 industry report, one-third of U.S. owners of a consumer physical activity monitor stopped using it within 6 months of having received it, while a little over 40% were still using it at after 24 months. The authors conclude: "Most of these devices fail to drive long-term sustained engagement for a majority of users" (39). The authors argue for three keys to sustained use: habit formation, social motivation, and goal reinforcement. Of course, there could easily be factors that would drive adherence differently—in both a positive or negative direction—in a research study as compared with the situation of consumers buying one for themselves or receiving one as a gift.

Comfort. As mentioned above, research-grade physical activity monitors are rarely used continuously for more than several weeks. The ability of consumer physical activity trackers to be used continuously for much longer periods is an advantage and also poses special challenges to researchers. Investigators need to consider factors that could interfere with long-term use of consumer physical activity monitors since these factors could impact participant adherence to the protocol. Some participants might not like wearing certain consumer physical activity monitors long-term because of stylistic concerns. Comfort level and potential physiological reactions may also be an issue. For example, according to ABC news, the Consumer Product Safety Commission has received more than 100 new complaints about a variety of consumer physical activity monitors over the past year due to what was dubbed "Fitbit rash" (http://abc7news.com/technology/7-on-your-side-rash-problem-remains-with-new-fitbit-trackers/512434/). Suggestions on how to clean devices to avoid skin irritations due to soap or sweat may need to be incorporated into a protocol.

Battery life. Many consumer physical activity monitors have limited battery life (typically, a week), which could be disadvantageous for research, as recharging may need to be done when meaningful activity is taking place. The need to recharge may also contribute to forgetting to wear the device. In contrast, by not having an energy-thirsty display or built-in optical HR monitor, some devices offer much longer battery life. Considering only the 308 wristband-type consumer physical activity monitors currently on the market, there are 33 that claimed a charge-free battery life of at least one year and one claiming two years (http://fitness-trackers.specout.com/). Even some of these long-lasting devices, however, may not have sufficient battery life for some longitudinal studies, in which case, an imperfect process of battery replacement could potentially cause an interruption in data collection.

Device placement. Physical activity monitors are most accurate if placed on the hip (54); however, many consumer devices that use accelerometer technology for step counting are made to be worn on the wrist (http://fitness-trackers.specout.com/). Some devices can be worn elsewhere, and some of these can be worn in a variety of ways. For example, the MisFit Shine can be worn on the wrist, as a pendant, in a pocket or clipped onto clothing, including the shoe. Users may find this flexibility desirable, but potential disadvantages for research include *1*) varying accuracy depending on the location on the body where the device is worn. Wrist-based devices are prone to misclassifying nonambulatory arm movements potentially confounding data analysis due to underestimation or overestimation of activity, as discussed above (50); and *2*) these flexibly worn devices may be more easily misplaced or forgotten than a device habitually worn on the wrist. On the other hand, devices that are flexibly worn and unobtrusive have the advantage of not competing with jewelry, fashionable watches, or wardrobes, and thus could potentially improve participant adherence.

User feedback. By collecting data on the device itself, the research-grade Actigraph GT9X Link & ActiLife (1a) allow an investigator to choose between enabling or disabling real-time subject feedback to the user that could impact outcome measures. Indeed, feedback about one's activity does impact the user (44), as discussed above under *Feedback from consumer physical activity monitors as a behavioral intervention to increase physical activity for disease prevention*, and this aspect needs to be taken into account in the study design.

Demographics. Another concern is the potential bias in the pool of participants, who have to own or have access to a smartphone and be interested in participating. Without addressing this selection bias, a sector of our population would not be represented, and results may not be generalizable to the entire population. Socioeconomic, gender, and racial disparities may impact participant recruitment due to not having a smartphone or access to one, which is frequently required for research using these consumer devices.

According to a 2015 Pew report (http://www.pewinternet.org/2015/10/29/the-demographics-of-device-ownership/), more than half of most demographic groups have a smartphone, and there are no differences in smartphone ownership among different racial and ethnic groups. There was significantly less smartphone ownership in individuals 65 and older (30% own smartphones) and those without a high school education (41% own smartphones). Furthermore, individuals living in rural communities had fewer smartphones than those living in suburban or urban communities. Less is known about the demographics of consumer physical activity monitors with respect to race. Thus, demographics pose potential study limitations, and it will be crucial to take demographics and the associated challenges into consideration in clinical study design, as well as recruitment and retention strategies.

Unique Opportunities and Challenges to Partner with Corporate and Healthcare Systems on Research Using Consumer Physical Activity Monitors

Opportunities. Consumer physical activity monitors are being used by companies and healthcare systems in conjunction with various behavioral motivation strategies to promote the health and wellness of their employees. Coca-Cola uses Misfit Shine to promote physical activity in their employees through challenges and rewards created by the Virgin Pulse tracking and engagement platform (https://misfit.com/health-wellness/#section-partners/). The community-based Passport Health Plan encourages employee physical activity by setting activity goals via monthly games that target the individual, as well as

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the whole company through the combination of Misfit Flash and Shine devices and the "Count.It" dashboard (https://misfit.com/health-wellness/#section-partners/). The Zurich-based company Dacadoo calculates a user's personal Health Score to motivate Misfit users through behavioral science and gaming techniques taking advantage of social networks and group dynamics (https://misfit.com/health-wellness/#section-partners/).

Other companies combine physical activity monitoring data with assessment of mood via a cloud-based data analytics platform. Koneksa Health goes even further and collects data on physical activity, mood, and biometrics, and links these data with environmental data, such as the weather and local complaints made over the state of the air, water, and sanitation (https://misfit.com/health-wellness/#section-work-with-us/). Location data are also collected, which enables factors like access to healthy food options to also be incorporated into research on patient health in the context of their environment.

While currently there are few studies identified in clinicaltrials.gov that involve partnerships between researchers and companies or healthcare systems, this number is likely to grow as consumer physical activity monitors become more widely used by the research community and as their technology improves. In one example, Duke is partnering with the Health Promotion Board in Singapore to conduct a randomized controlled trial of 950 full-time employees to determine whether economic incentives will promote walking among employees. The key outcome is changes in frequency of moderate to vigorous physical activity, which is measured by Fitbit Zip [NCT01855776].

Challenges. Bietz et al. (7) discusses at some length six major challenges for investigators using corporate data on the use of consumer physical activity monitors for health research. We briefly summarize them here.

1) Data ownership is often not clear. Many activity monitor companies have "click-through" agreements that cause the user to give away their personal information rights inadvertently to the company. This less-than-transparent approach can create distrust among participants, which could interfere with the integrity of human subject research.
2) Data access for research can be a challenge because Institutional Review Boards have to accept data collection practices of third parties, which are negotiated between corporate counsel and the researchers. Establishing standard language would help. Apple ResearchKit has made large-scale recruitment possible, if participants release their data for study purposes.
3) Privacy remains a concern, especially in this era of digital personal health. Privacy policies and practices for this kind of data that integrate seamlessly with those required by HIPAA regulations and other policies governing human subjects research will be key.
4) Informed consent and ethics need to be revisited, considering how they apply to new modes of research, and this presents an opportunity for new models and technologies of informed consent, deidentification, and trusted sharing.
5) Research methods and data quality are a concern because the data and methods are typically not shared by companies with the research community, and the reliability and validity of the equipment need to be verified.
6) The constantly evolving tech world of personal health presents new opportunities and challenges.

These limitations will be reduced as the technology improves, and there are great benefits from the "big data" these devices can provide. For example, Fitbit has a large data set (https://www.fitbit.com/fitscience/) on average user steps as a function of month, season, city, and even having a dog or not, all of which could inform the design of physical activity intervention trials. Clearly, there is much to be gained by the physiology research community in promoting partnerships between investigators and companies and healthcare systems that collect data on their employees, customers (e.g., Fitbit users), and patients.

Thus, there are challenges to overcome in the use of consumer physical activity monitors for health research, including data quality, given the reality of changing algorithms; data ownership, access, and privacy; informed consent and ethics; and other issues.

Opportunities to Use Smartphone Apps and Platforms Such as the Apple ResearchKit in Conjunction with Consumer Physical Activity Monitors

In March 2015, Apple released an open-source programming framework called ResearchKit (http://researchkit.org) that supports biomedical research via apps that take advantage of the capabilities of the iPhone or iWatch, including the accelerometer, gyroscope, and high-definition camera. Physical activity, HR, and sleep patterns are being followed in patients with chronic obstructive pulmonary disease using the StopCOPD app (15a) to determine their impact on respiratory symptoms. Patients with diabetes are being studied using the accelerometer and gyroscope in the iPhone to measure movement, along with self-reports of food intake and medication compliance through the GlucoSuccess app (43a). These data are being used to determine whether there are subtypes of Type 2 diabetes, which respond differently to exercise by studying the relationship between glucose levels and activity, diet, and treatment. The Concussion Tracker app (46a) is being used to investigate the impact of concussions on health by recording physical activity, HR patterns, and cognitive function for six weeks after a head injury. Combining the physical activity monitor capability with the GPS feature in the iPhone could also enable research on exercise and air quality in people with asthma using the Asthma Health app (26a), which could not only provide important research findings but also enable discovery of adverse behavior and environments for individuals with asthma.

One of the major advantages of using ResearchKit is the ability to recruit human subjects throughout the world. Over 10,000 participants with Parkinson's disease have enrolled in a study on the association of exercise, sleep, and mood with measures of dexterity, balance, gait, and memory using the mPower app (10). The ability to recruit large populations is helping investigators who are attempting to develop algorithms that predict an event, in some cases even personalized events. For example, the EpiWatch app (33a) developed by the Johns Hopkins Epilepsy Center uses the accelerometer and HR sensor on the iWatch to determine whether a personal algorithm can be constructed for predicting seizures in patients with epilepsy (http://www.hopkinsmedicine.org/epiwatch).

Apple has recently launched a new framework called CareKit (http://carekit.org/) that allows developers to build apps around patient care. The Insight Dashboard module enables tracking of patient progress, which can be used to investigate the effectiveness of treatments. The Care Card module of CareKit (2a) tracks pain, temperature, hunger, dizziness, range of mobility, and medication and by being connected to medical teams, enables medical specialists to send alerts to patients, family members, or caregivers in case of adverse events. For example, if a patient's normal level of physical activity were to fall below a certain threshold, alerts could be sent to check on the patient. When coupled with the One Drop app (30), glucose levels are incorporated to improve management of blood sugar in patients with diabetes.

Thus, not only are these platforms enabling earlier diagnosis and treatment, they also are providing powerful opportunities for research, including in recruitment strategies, big data analyses, identification of and cost-effective inclusion of additional salient variables, and understanding personal differences.

Next-Generation of Consumer Physical Activity Monitors

Pattern recognition. While consumer technology is beginning to differentiate walking from other activities (e.g., Fitbit says its algo-

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rithms, when enabled by the user, are also able to recognize runs, bike rides, elliptical, sports, aerobic workouts, and swims) (https://blog.fitbit.com/introducing-fitbit-flex-2-the-fitness-tracker-that-fits-your-life/), the technology has not reached the point where it can differentiate other forms of physical activity from each other very well. Thus, the energy expenditure exerted during cycling does not receive sufficient credit compared with walking (17). Accordingly, a consumer physical activity monitor worn on the wrist of a ballroom dancer (particularly when doing the Viennese Waltz well) would grossly underestimate the number of steps. Continued advances in pattern recognition to identify specific physical activities will allow classification of physical activity modes, and provide contextual information (location and setting). In this regard, recognition of resistance training, which is poorly captured currently by both consumer and research-grade physical activity monitors, is an important technological goal because of its major contribution to a complete physical fitness plan (37). Through improved pattern recognition, calculations of energy expenditure will become more accurate for each activity, enabling comparisons among activities (i.e., how similar is surfing to skiing?) and providing consolidated energy expenditures for the individual.

Pattern recognition will also enable studies investigating how the "type" of physical activity impacts physiology and health outcomes. For example, we and others hypothesize that complex exercise is more effective than simple exercise for cognitive preservation. This hypothesis is based on animal studies showing combined exposure of physical and cognitive activities is superior to maintaining or improving working memory than either activity alone (38). Human studies also support this concept. A study of leisure activities and the risk of dementia in the elderly found that the physical activity associated with the lowest hazard ratio for dementia was dancing, which was a leisure activity in their physical activities category that had a substantial cognitive component. In fact, the hazard ratio for dancing was significantly lower than walking or swimming (which have less cognitive contributions), or for cognitive activities alone like crossword puzzles or reading (60). Thus, the ability of consumer physical activity monitors to recognize and then accurately compare energy expenditure in dancers vs. swimmers will help address this intriguing hypothesis.

Integration of consumer physical activity monitors with sleep monitors. Many consumer physical activity monitors now include assessments of sleep, and this feature is attracting the interest of the behavioral sleep medicine field. The Withings Activité watch reports daily on when you went to bed, how long it took you to fall asleep, how long you were in bed, how long you were asleep, when you awoke and when you rose from bed (https://www.withings.com/us/en/products/activite/). The device also divides the sleep period into light- and deep-sleep intervals, and it provides a summary of the length of time spent in light and deep sleep, as well as the number of times you awoke during the sleep period. Fitbit, Misfit, Garmin, and Nike, as well as many other producers of consumer devices provide similar data.

Assessment of sleep is a natural extension of the technology used to measure physical activity given that sleep patterns can be monitored using algorithms based on accelerometry data collected through wearable devices (45). There is a growing body of literature examining the validity and reliability of various consumer sleep-monitoring devices with outcomes dependent on the type of device and the sleep parameters that are being assessed. Sleep is complicated, however, and provides significant challenges to the current technology (48). The absence of movement is not necessarily equated with a sleep period, as it is common for individuals to move during the night and for individuals to be sedentary during the day. Automatically distinguishing between naps, sleep, and nonwearing times needs to improve (9). Various algorithms are currently being developed to increase accuracy based on extended periods of physical inactivity, consistent increases in movement following a prolonged period of inactivity, arm angle relative to a horizontal plane, wrist rotations, gravitational pull, and episodic sampling techniques that influence sensitivity (26, 31, 32, 59).

Consumer devices measuring sleep have not yet achieved the same level of accuracy as consumer physical activity monitors have in counting steps, and significant flaws in sleep assessments remain. A few studies have examined the accuracy of the devices, which have generally been found to be accurate compared with polysomnography recordings (the gold standard for sleep assessments) on total sleep time but less so for measures, such as identifying sleep interruptions (19, 43). Despite overall agreement with polysomnography, the Jawbone Up overestimated total sleep time and sleep onset latency, while wake after sleep onset was underestimated, especially on nights with greater nocturnal arousals (18). As with calculations of physical activity, there is no single algorithm used by all consumer devices to assess sleep. This lack of consensus contributes to measurement inconsistencies among devices and across sleep parameters.

Integration of consumer physical activity monitors with other biosensors. The field of wearable biosensors is exploding, and the interconnectedness and integration of various kinds of consumer and research-grade biosensors in conjunction with consumer physical activity monitors has the potential to provide even greater insights in biomedical research than these devices can in isolation, enabling a more comprehensive analysis of the ecological assessment of human physiology in biomedical research. Biosensors measuring HR are the most recent addition to the consumer physical activity device industry. Thus far, only a few studies have reported on the validity of HR measures. One study compared several commercial activity HR monitors against the Onyx Vantage 9590 clinical pulse oximeter and found that accuracies ranged between 80% and 99%, and the precision ranged between 4% and 18%; MisFit Shine exhibited the highest accuracy and precision, while Samsung Gear 2 had the poorest accuracy and Jawbone Up had the poorest precision (24). Another study comparing HR measured by Basis Peak and Fitbit Charge against a 12-lead electrocardiograph system showed accuracies were reduced with increasing exercise intensity for both commercial activity monitors; however, the Fitbit charge was far less accurate than the Basis Peak at high-intensity levels, presumably because of factors (e.g., ambient light, sweat, skin contact force) that interfere with the biosensor (33). In fact, Fitbit is currently facing lawsuits regarding the accuracy of the HR monitors on the Fitbit Charge HR and Surge devices (which currently use LED lights and optical sensors to detect changes in capillary blood volume, and then use algorithms to calculate HR) (2).

Wearable biosensors are also being developed for measuring blood pressure without the bulky cuffs that measure blood pressure through the brachial artery (https://www.wearable-technologies.com/2016/05/the-silent-killer/). Omron is currently seeking regulatory clearance from the Food and Drug Administration for its wrist blood pressure monitor termed Project Zero (https://www.youtube.com/watch?v=BmO3SAy69Hw/), which looks much more like a large commercial activity monitor than the peer-reviewed Omron R2 (58), which although worn on the wrist still looks like a mini-upper arm blood pressure-monitoring device. H2 made by Happiness by Health (https://www.youtube.com/watch?v=bglVYynr0dQ/) is the smallest wrist worn device reported to be in development and is using crowd sourcing to fund completion of the project. The Salu Pulse Band made by a Canadian startup estimates ambulatory blood pressure via blood flow around the wrist. Transdermal biosensors in the form of a patch or embedded in clothes just adds to the possibilities, especially those that can sample body fluid components like glucose, lactate, or proteins or that monitor blood alcohol levels or allergens in the air (41).

Information processing. Computing tools aid researchers in a multitude of tasks from solving complex nonlinear equations to specific applications like IBM's Watson supercomputer. The connecting of computing devices (the Internet, WiFi, Bluetooth) has become

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so common that we sometimes forget about their role in facilitating scientific advancement. As we enter the era of the "Internet of things," Internet-connected sensor-based technologies such as consumer physical activity monitors are new tools with the potential to enable breakthroughs in our understanding and promotion of human health.

Perspectives and Significance

Consumer physical activity monitors have already transformed physiology research by enabling investigators to continuously measure physical activity for months if not years in an economical manner, and—importantly—using vehicles that are popular and widely available wearable devices. The wealth of information offered by these platforms may enable researchers to assess physical activity in new ways, approximating free-living conditions like never before, on a new level of ecological validity that may lead to new, unanticipated discoveries (14). This is true of both naturalistic studies and randomized control trials.

Scientific discovery has often made advances after new tools for investigation became available. This has included tools that extend the range of our human senses like microscopes, telescopes, and X-rays, as well as sophisticated tools like particle colliders, PET/CT scanners, and MRI machines. The combination of new technological tools with information processing breakthroughs is revolutionizing many areas of research. Developments in areas that may open new possibilities for research are already in the pipeline. Gains in accuracy, in the effectiveness of algorithms, and in the interconnectedness and integration of various kinds of sensors and measurements, have the potential not only to provide incremental gains, but at some point to engender sudden leaps in what can be learned. Real-time coordination and gating of information inputs may sometimes have a radically synergistic effect. The current state and potential growth of this technology are transforming physiology research and are enabling us to ask new and more granular questions about activity and sleep in health and disease. These are exciting times.

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No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

S.P.W. conceived and designed research; S.P.W. and K.S. analyzed data; S.P.W., T.S.H.B., S.R.C., and K.S. drafted manuscript; S.P.W., T.S.H.B., S.R.C., and K.S. edited and revised manuscript; S.P.W., T.S.H.B., S.R.C., and K.S. approved final version of manuscript; S.P.W. and K.S. prepared figures.

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